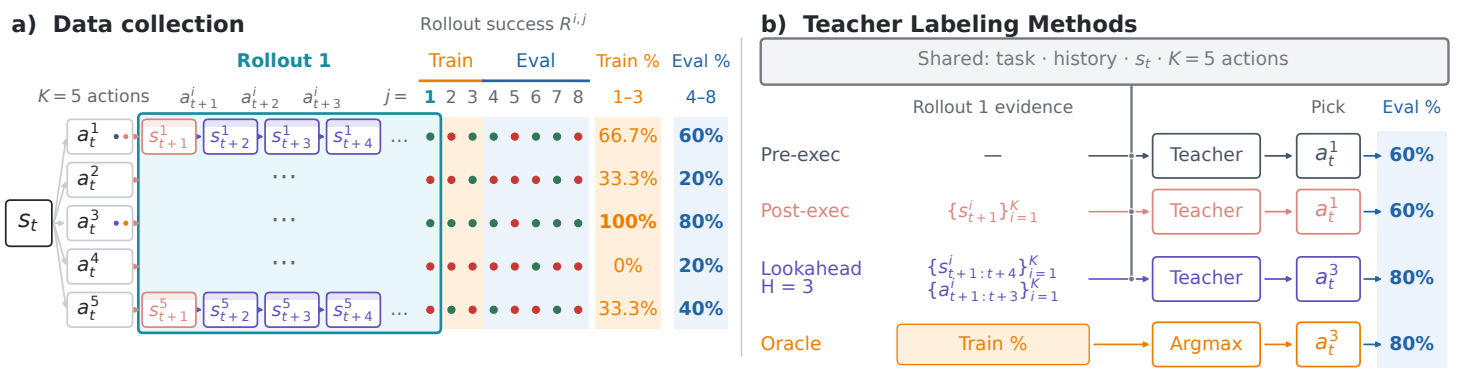
a) Data collection
Rollout success $R^{i,j}$
Rollout 1
Train
Eval
Train %
Eval %
$K = 5$ actions
a^i_{t+1}
a^i_{t+2}
a^i_{t+3}
$j =$ 1 2 3 4 5 6 7 8
1–3
4–8
s_t
a^1_t
a^2_t
a^3_t
a^4_t
a^5_t
s^1_{t+1} s^1_{t+2} s^1_{t+3} s^1_{t+4} ...
s^5_{t+1} s^5_{t+2} s^5_{t+3} s^5_{t+4} ...
66.7%
33.3%
100%
0%
33.3%
60%
20%
80%
20%
40%
b) Teacher Labeling Methods
Shared: task · history · s_t · $K = 5$ actions
Rollout 1 evidence
Pick
Eval %
Pre-exec
—
Post-exec
$\{s^i_{t+1}\}^K_{i=1}$
Lookahead H = 3
$\{s^i_{t+1:t+4}\}^K_{i=1}$
$\{a^i_{t+1:t+3}\}^K_{i=1}$
Oracle
Train %
Teacher
Argmax
a^1_t
a^3_t
60%
60%
80%
80%
Success
Failure
Pre-exec
Post-exec
Lookahead H = 3
Oracle